SALES PERFORMANCE DASHBOARD

Comprehensive Data Analytics & Business Intelligence Report

Prepared by:
Cynthia Ikhiannosen OGBEKHIULU
Data Analytics Intern, Syntecxhub

Date: May 2026

Contents

1. Executive Summary

This project report outlines the structural framework, methodologies, and core execution phases involved in creating the Syntecxhub Sales Performance Dashboard. The goal of this implementation is to transform traditional raw transactional datasets into dynamic, high-impact business intelligence assets. By automating data cleaning pipelines, engineering optimized analytical models, and providing granular visual breakdown capabilities, this dashboard provides decision-makers with real-time strategic clarity over corporate revenue channels.

> *Key Insight: Navigating real-world technical data operations often introduces integration complexities. Despite unexpected procedural challenges experienced with Git version control and GitHub remote hosting protocols during delivery, the full pipeline architecture was successfully staged, committed, and pushed to production repositories.*

2. Core Project Objectives

Adhering strictly to the technical framework assigned under the Syntecxhub Data Analytics initiative, the lifecycle of this project was successfully executed across six primary functional milestones:

- **Dataset Ingestion & Quality Assurance:** Importing raw multi-channel transactional tables into Power BI Desktop, followed by rigorous operational data cleaning via Power Query.
- **Temporal Trend Mapping:** Analyzing systemic sales momentum variations across monthly, quarterly, and fiscal year horizons to catch seasonal trends.
- **Inventory & Product Diagnostics:** Isolating critical performance attributes to precisely rank top-selling products alongside low-performing baseline stock lines.
- **Geographical & Segment Comparison:** Executing detailed matrix evaluations on cross-regional revenues and macro product category behaviors.
- **Strategic KPI Engineering:** Developing dynamic, performant Data Analysis Expressions (DAX) to monitor corporate vital metrics including Total Revenue, Profit, and Growth Rates.
- **Interactive Interface Design:** Structuring an executive-ready, highly intuitive interactive visualization layout designed for immediate exploratory data analysis.

3. Data Preprocessing & Power Query Pipelines

Data reliability was established at the source by executing an intensive Extract-Transform-Load (ETL) pipeline inside the Power Query engine before analytical formulas were layered. The schema cleanup included:

- **Structural Cleansing:** Systematic evaluation and removal of null records, missing cell targets, and structural duplicates to guarantee mathematical correctness across aggregated dimensions.
- **Data Type Normalization:** Enforcing explicit database schema constraints (e.g., configuring text attributes, standardizing currency metrics, and transforming transactional timestamps into uniform ISO-compliant date formats).
- **Dimensional Enrichment:** Structuring foundational relational logic to allow flawless filtering down across continuous time-series charts, item catalogs, and sales territories.

4. Analytical Modeling & DAX Metric Formulas

To support instant dashboard responsiveness and scalable aggregations, key corporate performance fields were built via advanced Data Analysis Expressions (DAX):

- **Total Revenue:** Aggregated via iterative line-level transaction evaluations multiplying unit sales volumes against retail price fields.
- **Net Corporate Profit:** Calculated by subtracting exact manufacturing cost bases and operational cost metrics from gross revenue values.
- **Compounded Growth Rate:** Time-intelligence metrics tracking sequential trend momentum by dynamically contrasting current period values against historical baseline benchmarks.

5. Visual Architecture & User Guide

The front-end user experience uses a professional corporate deep slate color palette (#0F172A) contrasted against sharp accent tones for intuitive reading:

- **Executive KPI Scorecards:** Placed at the top of the interface for immediate situational awareness regarding high-level strategic financial outputs.
- **Interactive Filter Matrix:** Equipped with fluid slice-and-dice selectors allowing managers to dynamically segment operational reports by unique distribution channels (In-store, Online, and Outlet formats).
- **Geographical Drill-Down Maps:** Enabling immediate geographical audits across various regions to proactively identify underperforming regional distribution nodes.

6. Repository & Version Control Specifications

The entire source control logic, metadata definitions, background graphic configurations, and visual assets are securely cataloged. Production files can be inspected via the official version control tree linked below:

Official GitHub Repository:

https://github.com/CynthiaOgbe2907/Syntecxhub_Sales.Performance.Dashboard.Project_Cynthia.OGBEKHIULU